



NVAPI Open Source SDK for Driver Release 610

Release Notes

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NVAPI Release Notes

Introduction

NVAPI is NVIDIA Corporation's core software development kit that allows direct access to NVIDIA GPUs and drivers on all Windows platforms. NVAPI provides support for categories of operations that range beyond the scope of those found in familiar graphics APIs such as DirectX and OpenGL.

The following files are provided by NVIDIA:

- > nvapi.h
- > nvapi_interface.h
- > nvapi_lite_common.h
- > nvapi_lite_d3dext.h
- > nvapi_lite_salend.h
- > nvapi_lite_salstart.h
- > nvapi_lite_sli.h
- > nvapi_lite_stereo.h
- > nvapi_lite_surround.h
- > NvApiDriverSettings.c
- > NvApiDriverSettings.h
- > nvHLSLExtns.h
- > nvHLSLExtnsInternal.h
- > nvShaderExtnEnums.h
- > \x86\nvapi.lib
- > \amd64\nvapi64.lib
- > \docs\NVAPI_Reference_OpenSource.chm
- > \docs\NVAPI_SDKs_Samples_and_Tools_License_Agreement(Public).pdf

These release notes describe the new features, enhancements, and changes in the NVAPI SDK for this release.

Changes in NVAPI for Driver Release 610

New Functions

- > Added `NvAPI_GPU_GetOverclockStatus`
- > Added `NvAPI_GPU_NVLINK_GetCapsEx`
- > Added `NvAPI_GPU_NVLINK_GetStatusEx`

New/Updated Structures

- > Added `NV_GPU_OVERCLOCK_STATUS_V1`
- > Added `NVAPI_NVLINK_LINK_MASK_V1`
- > Added `NVLINK_GET_CAPS_EX_V1`
- > Added `NVLINK_GET_STATUS_EX_V1`
- > Added `NV_GSYNC_DISPLAY_V1`
- > Added `NV_GSYNC_DISPLAY_V2`
- > Added `u64CurrentValue` to `NVDRS_SETTING_V1`
- > Added `u64PredefinedValue` to `NVDRS_SETTING_V1`
- > Added `u64DefaultValue` to `NVDRS_SETTING_VALUES`
- > Added `u64Value` to `NVDRS_SETTING_VALUES`

New/Updated Enums

- > Added the following enum to `NVAPI_RAY_FLAG_EX`
 - o `NVAPI_RAY_FLAG_SKIP_BUILTIN_PRIMITIVES_EX`
- > Added the following enum to `NVDRS_SETTING_TYPE`
 - o `NVDRS_QWORD_TYPE`
- > Added the following enum to `NV_NGX_DRIVER_FEATURE_ID`
 - o `NV_NGX_DRIVER_FEATURE_ID_FRAME_PRESENT_NOTIFY_HYBRID`

New Typedefs

- > Added `NVLINK_GET_CAPS_EX`
- > Added `NVLINK_GET_STATUS_EX`
- > Added `NV_GSYNC_DISPLAY`

New Unions

- > None

New Macros

- > Added NVAPI_NVLINK_MAX_LINKS_V2
- > Added NVLINK_GET_CAPS_EX_VER
- > Added NVLINK_GET_CAPS_EX_VER1
- > Added NVLINK_GET_STATUS_EX_VER
- > Added NVLINK_GET_STATUS_EX_VER1
- > Added NV_GPU_OVERCLOCK_STATUS_VER
- > Added NV_GPU_OVERCLOCK_STATUS_VER1
- > Added NV_GSYNC_DISPLAY_VER1
- > Added NV_GSYNC_DISPLAY_VER2
- > Added NV_GSYNC_GPU_VER

New Errors

- > None

TCC Support

- > None

MCDM Support

- > None

NVAPI Security Information

- > None

Deprecated NVAPI Functions

- > None

Deprecated Enum Values

- > None

NVAPIDriverSettings Additions/Removals

New DRS Macros

- > Added `NGX_DLSS_NR_OVERRIDE_RENDER_PRESET_SELECTION_STRING`
- > Added `NGX_DLSS_NR_OVERRIDE_STRING`
- > Added `NGX_DLSS_NR_SL_OVERRIDE_STRING`

New DRS Enums

- > Added the following enums to `ESetting`
 - o `NGX_DLSS_NR_OVERRIDE_ID`
 - o `NGX_DLSS_NR_OVERRIDE_RENDER_PRESET_SELECTION_ID`
 - o `NGX_DLSS_NR_SL_OVERRIDE_ID`
- > Added `EValues_NGX_DLSS_NR_OVERRIDE`
- > Added `EValues_NGX_DLSS_NR_OVERRIDE_RENDER_PRESET_SELECTION`
- > Added `EValues_NGX_DLSS_NR_SL_OVERRIDE`

New DRS Value Arrays (.c)

- > Added `g_valuesNGX_DLSS_NR_OVERRIDE`
- > Added `g_valuesNGX_DLSS_NR_OVERRIDE_RENDER_PRESET_SELECTION`
- > Added `g_valuesNGX_DLSS_NR_SL_OVERRIDE`

New DRS Settings (.c)

- > Added `NGX_DLSS_NR_OVERRIDE_ID`
- > Added `NGX_DLSS_NR_OVERRIDE_RENDER_PRESET_SELECTION_ID`
- > Added `NGX_DLSS_NR_SL_OVERRIDE_ID`

HLSL Extension Additions/Removals

- > None

NVAPI Security Information

User administrator privilege is required to access certain driver features per NVIDIA's overall security vision. This helps mitigate the impact of malware.

Each API requiring administrator access will return an `NVAPI_INVALID_USER_PRIVILEGE` error, when run with standard user privilege.

The application will require administrator privileges to access this API, which can be elevated to a higher permission level by selecting "Run as Administrator" in Admin approval mode.

The following API requires administrator privileges:

```
>  NvAPI_GPU_SetEDID
```

Sample Code

The SDK package contains the `Sample_Code` directory, which provides code examples for the following features:

Feature	Sample Code Subdirectory	Sample Code
Custom timing	CustomTiming	CustomTiming.cpp
Display color control	DisplayColorControl	> DisplayColorControl.cpp > NVHelper.cpp > NVHelper.h > stdafx.h > targetver.h
Display configuration	DisplayConfiguration	> DisplayConfiguration.cpp > targetver.h
GPU handle enumeration	GPUHandleEnumeration	gpuHandleEnumeration.c
QSYNCEvent registration	QSYNCEventRegistration	QSYNCEventRegistration.cpp
Sync configuration	Sync_Configuration	> Sync_Configuration.cpp > targetver.h

I2C	i2c	> i2c.cpp > targetver.h
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